

Total No. of Printed Pages: 2

SUBJECT CODE NO: - RNP-8060
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group B) (NEP) Sem - II
Examination May/June-2026
Basics of Electronics Engineering

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

1. Part A: Question No 1 is compulsory.
2. Attempt any four questions from Part B: Q.2 to Q.7.
3. Use of non-programmable calculator is allowed.

Part A**Q.1 Solve the following questions****20**

- 1) Draw the symbols of Zener Diode, LED, BJTs & FETs
- 2) Which of the following statements is correct?
 - a) AC has fixed Polarity & 50 Hz Frequency
 - b) DC has fixed Polarity & Zero Frequency
 - c) DC & AC both has Polarity & 50 Hz Frequency
 - d) Frequency of AC & DC = 0.
- 3) Define HLR & VLR in Mobile Communication System
- 4) Convert $(1100.1011)_2$ this Binary number into its equivalent Decimal number:
- 5) Match the pairs:

Sr. No.	Value of CFR Resistor	Color Codes
1	Brown, Black, Yellow, Brown	Red, White, Green, Silver
2	120 K Ω with 10% Tolerance	100 K Ω with 1% Tolerance
3	2.9M Ω with 10% Tolerance	Brown, Black, Gold, Gold
4	1 Ω with 5% Tolerance	Brown, Red, Yellow, Silver

6) Perform following operation on given Binary number:

$$1011 + 1011 + 1100 + 1111 = ?$$

7) Op-Amp can amplify:

- | | |
|----------------------|----------------------------|
| (i) Only AC signals | (iii) Both AC & DC signals |
| (ii) Only DC signals | (iv) Noise signals |

8) The voltage gain of inverting amplifier is:

- | | |
|----------------------|----------------------|
| (iii) R_f / R_{in} | (iii) R_{in} / R_f |
| (iv) $-R_f / R_{in}$ | (iv) 1 |

- 9) **A transducer converts:**
- Electrical energy to mechanical energy
 - Mechanical energy to electrical energy
 - One form of energy into another
 - Heat energy into light
- 10) **RTD works on the principle of:**
- Change in capacitance
 - Change in resistance with temperature
 - Change in inductance
 - Piezoelectric effect

Part B

Attempt ANY FOUR questions from Part B: Q.2 to Q.7.

- | | | |
|------------|---|-----------|
| Q.2 | (a) What is a Rectifier? State its types & explain any one with suitable diagrams | 05 |
| | (b) Explain the Block diagram of DC Regulated Power Supply | 05 |
| Q.3 | (a) Define BJT with types, structure & symbol. Explain its working | 05 |
| | (b) Define FET with types, structure & symbol. Explain its working | 05 |
| Q.4 | (a) Define Op-Amp? Draw & Explain the Block diagram of Op-Amp | 05 |
| | (b) Explain a summing Amplifier with its circuit Diagrams | 05 |
| Q.5 | (a) What are different Number Systems with suitable examples | 05 |
| | (b) Explain the Logic Gates with symbols & Truth Tables | 05 |
| Q.6 | (a) Define a Transducers, state its classification with examples | 05 |
| | (b) Explain the working of RTD. State its advantages & disadvantages | 05 |
| Q.7 | (a) Draw & Explain all the elements of a communication System | 05 |
| | (b) Define following terminologies used in Mobile communication | 05 |
| | (i) BTS | |
| | (ii) HLR | |
| | (iii) VLR | |
| | (iv) EIR | |
| | (v) AUC | |